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[illegible]



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Head of the
Department

External
Examiner

Internal Examiner
Subject teacher

Date : / / 20 Department of



PRACTICAL 1: LOAD CSV, INSPECT, HANDLE MISSING VALUES

Code –

```
import pandas as pd

df = pd.read_csv("sales_data.csv")

print(df.tail(10))

print(df.dtypes)

print(df.isnull().sum())

num_cols = df.select_dtypes(include="number").columns

df[num_cols] = df[num_cols].fillna(df[num_cols].mean())

print(df.isnull().sum())
```

Output :

```
PS O:\vip> & "C:/Program Files/Python313/python.exe" o:/vip/1.py
      date      product  category  price  quantity  revenue
359  2022-12-22   Sneakers   Shoeses   80.0      12.0    960.0
360  2022-12-23    T-Shirt  Clothing   20.0      25.0    500.0
361  2022-12-24  Smartwatch  Accessories 200.0       8.0   1600.0
362  2022-12-25     Jeans   Clothing   50.0      15.0    750.0
363  2022-12-26  Smartphone  Electronics 600.0       9.0   5400.0
364  2022-12-27     Watch  Accessories 150.0       5.0    750.0
365  2022-12-28     Coat   Clothing  100.0       5.0    500.0
366  2022-12-29  Headphones  Electronics 100.0      10.0   1000.0
367  2022-12-30  Smartphone  Electronics 600.0      11.0   6600.0
368  2022-12-31    Hoodie   Clothing   40.0      30.0   1200.0
date      str
product    str
category    str
price      float64
quantity   float64
revenue     float64
dtype: object
date      0
product    0
category    0
price      2
quantity    1
revenue     1
dtype: int64
date      0
product    0
category    0
price      0
quantity    0
revenue     0
```

PRACTICAL 2: COLUMN-WISE OPERATIONS

```
import pandas as pd

df = pd.read_csv("sales_data.csv")

df2 = pd.read_csv("employee_data.csv")

print(df.head())

print(df2.head())

df.rename(columns={
    'product': 'product1',
    'age': 'age2'
}, inplace=True)

df['Profit'] = df['price'] * df['revenue']

df.rename(columns={
    'Profit': 'Calculated_Value'
}, inplace=True)

print(df.head())

print(df2.head())
```

```
PS O:\vip> & "C:/Program Files/Python313/python.exe" o:/vip/2.py
   date  product  category  price  quantity  revenue
0 2022-01-01  Smartphone  Electronics  600.0      10.0    6000.0
1 2022-01-01   Laptop  Electronics 1200.0       5.0    6000.0
2 2022-01-02   T-Shirt   Clothing    20.0     50.0   1000.0
3 2022-01-03  Headphones  Electronics  100.0     20.0   2000.0
4 2022-01-04   T-Shirt   Clothing    20.0     25.0    500.0
  department  salary  age
0     Sales  110390   41
1   Finance   92329   29
2        IT   83088   64
3     Sales   79811   33
4  Marketing   95479   22
   date  product1  category  price  quantity  revenue  Calculated_Value
0 2022-01-01  Smartphone  Electronics  600.0      10.0    6000.0    3600000.0
1 2022-01-01   Laptop  Electronics 1200.0       5.0    6000.0    7200000.0
2 2022-01-02   T-Shirt   Clothing    20.0     50.0   1000.0     20000.0
3 2022-01-03  Headphones  Electronics  100.0     20.0   2000.0    200000.0
4 2022-01-04   T-Shirt   Clothing    20.0     25.0    500.0     10000.0
  department  salary  age
0     Sales  110390   41
1   Finance   92329   29
2        IT   83088   64
3     Sales   79811   33
4  Marketing   95479   22
```

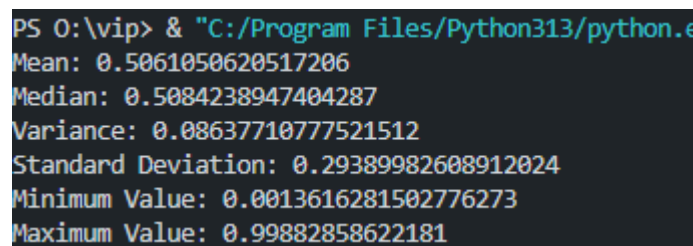
PRACTICAL 3: NUMPY STATISTICS ON RANDOM FLOATS

```
import numpy as np
```

```

data = np.random.rand(500)
mean = np.mean(data)
median = np.median(data)
variance = np.var(data)
std_dev = np.std(data)
min_val = np.min(data)
max_val = np.max(data)
print("Mean:", mean)
print("Median:", median)
print("Variance:", variance)
print("Standard Deviation:", std_dev)
print("Minimum Value:", min_val)
print("Maximum Value:", max_val)

```



```

PS 0:\vip> & "C:/Program Files/Python313/python.e
Mean: 0.5061050620517206
Median: 0.5084238947404287
Variance: 0.08637710777521512
Standard Deviation: 0.29389982608912024
Minimum Value: 0.0013616281502776273
Maximum Value: 0.99882858622181

```

PRACTICAL 4: GROUP BY TWO CATEGORICAL COLUMNS

Code -

```

import pandas as pd
df2 = pd.read_csv("employee_data.csv")
grouped = df2.groupby(["department"])[["salary"]].agg(
    count="count",
    total="sum",
    average="mean"
)
print(grouped)

```

Output –

	count	total	average
department			
Engineering	156	12384319	79386.660256
Finance	156	12869371	82495.967949
HR	137	10950290	79929.124088
IT	148	11434788	77262.081081
Marketing	130	10204708	78497.753846
Operations	135	10705424	79299.437037
Sales	138	10986798	79614.478261

PRACTICAL 5: FILTERING USING .LOC() AND .QUERY()

```
import pandas as pd

df2 = pd.read_csv("employee_data.csv")

grouped = df2.groupby(["department"])["salary"].agg(
    count="count",
    total="sum",
    average="mean"
)

print(grouped)

filtered_loc = df2.loc[(df2["age"] > 25) & (df2["salary"] < 50000)]

filtered_query = df2.query("age > 25 and salary < 50000")

print(filtered_loc.head())

print(filtered_query.head())
```

```

PS O:\vip> & "C:/Program Files/Python313/python.exe" o:/vip
count      total      average
department
Engineering 156 12384319 79386.660256
Finance      156 12869371 82495.967949
HR           137 10950290 79929.124088
IT           148 11434788 77262.081081
Marketing    130 10204708 78497.753846
Operations   135 10705424 79299.437037
Sales        138 10986798 79614.478261
  department salary age
9    Marketing 45569 58
14   Finance  46254 27
18   HR       46090 30
42   Sales    45240 59
48   Engineering 44432 42
  department salary age
9    Marketing 45569 58
14   Finance  46254 27
18   HR       46090 30
42   Sales    45240 59
48   Engineering 44432 42

```

PRACTICAL 6: NUMPY MATRIX OPERATIONS

Code –

```
import numpy as np
```

```
A = np.random.randint(1, 10, (4, 4))
```

```
B = np.random.randint(1, 10, (4, 4))
```

```
addition = A + B
```

```
subtraction = A - B
```

```
multiplication = np.matmul(A, B)
```

```
transpose = A.T
```

```
determinant = np.linalg.det(A)
```

```
print("Addition:")
```

```
print(addition)
```

```
print("\nSubtraction:")
```

```
print(subtraction)
```

```
print("\nMultiplication:")
```

```
print(multiplication)
```

```
print("\nTranspose:")
```

```
print(transpose)

print("\nDeterminant:")

print(determinant)
```

```
PS O:\vip> & "C:/Program Files/Python313/python.exe"
Addition:
[[11 16  7  6]
 [ 4 10  5 14]
 [ 3 15  8  6]
 [15  6 12  8]]

Subtraction:
[[-5 -2 -3  2]
 [ 0  6 -3  0]
 [ 1 -3  2  4]
 [-3 -4 -6 -2]]

Multiplication:
[[ 76  79  85  77]
 [ 96  78 108  96]
 [ 78 100  94  76]
 [ 80  98  70  37]]

Transpose:
[[3 2 2 6]
 [7 8 6 1]
 [2 1 5 3]
 [4 7 5 3]]

Determinant:
466.00000000000003
```

PRACTICAL 7: MERGE THREE DATAFRAMES

```
import pandas as pd

customers = pd.DataFrame({
    "customer_id": [1, 2, 3],
    "customer_name": ["Amit", "Sara", "John"]
})

products = pd.DataFrame({
    "product_id": [101, 102],
    "product_name": ["Laptop", "Phone"]
})
```



```

}))

orders = pd.DataFrame({
    "order_id": [1, 2, 3],
    "customer_id": [1, 2, 1],
    "product_id": [101, 102, 102],
    "amount": [50000, 20000, 22000]
})

merged = orders.merge(customers, on="customer_id") \
    .merge(products, on="product_id")

print(merged)

```

Output –

```

PS O:\vip> & "C:/Program Files/Python313/python.exe" o:/vip/7.py

```

	order_id	customer_id	product_id	amount	customer_name	product_name
0	1	1	101	50000	Amit	Laptop
1	2	2	102	20000	Sara	Phone
2	3	1	102	22000	Amit	Phone

PRACTICAL 8: DATE TIME OPERATIONS

Code –

```

import pandas as pd

data = {
    'Date': ['2023-5-1', '2023-6-10', '2023-7-12', '2023-8-14', '2023-9-10'],
    'Sale': [600, 900, 800, 1200, 500]
}

df = pd.DataFrame(data)

df['Date'] = pd.to_datetime(df['Date'], format='mixed')

df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['Day'] = df['Date'].dt.day
df['Day_Name'] = df['Date'].dt.day_name()

print(df)

```

```
PS 0:\vip> & "C:/Program Files/Python313/python.exe"
      Date  Sale  Year  Month  Day  Day_Name
0 2023-05-01   600  2023     5    1   Monday
1 2023-06-10   900  2023     6   10  Saturday
2 2023-07-12   800  2023     7   12  Wednesday
3 2023-08-14  1200  2023     8   14   Monday
4 2023-09-10   500  2023     9   10   Sunday
```

PRACTICAL 9: HISTOGRAM PLOT

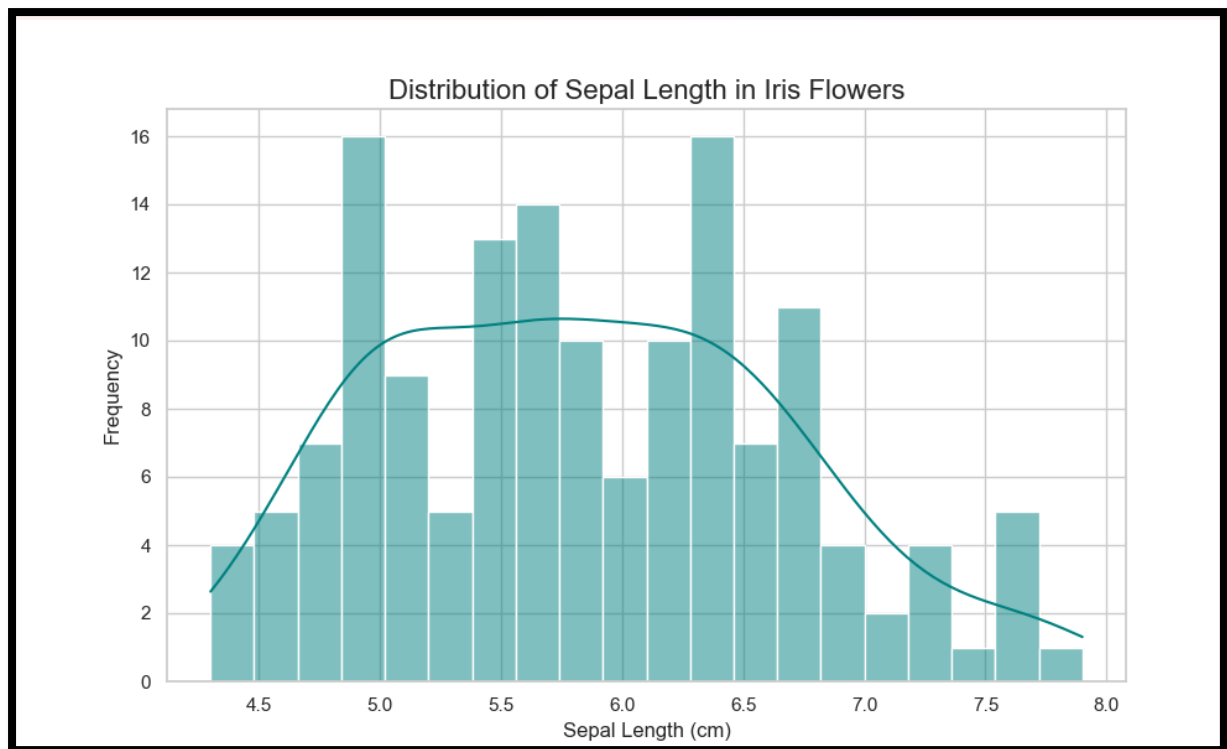
Code –

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('iris')
sns.set_theme(style="whitegrid")
plt.figure(figsize=(10, 6))

plot = sns.histplot(
    data=df,
    x='sepal_length',
    bins=20,
    color='teal',
    kde=True
)

plt.title('Distribution of Sepal Length in Iris Flowers', fontsize=16)
plt.xlabel('Sepal Length (cm)', fontsize=12)
plt.ylabel('Frequency', fontsize=12)
plt.show()
```



PRACTICAL 10: HEATMAP PLOT

Code –

```
from sklearn.datasets import load_iris
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

iris_data = load_iris()
iris_df = pd.DataFrame(data=iris_data.data, columns=iris_data.feature_names)
corr_matrix = iris_df.corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix,
            annot=True,
            fmt=".2f",
            cmap='YlGnBu',
            linewidths=.5)

plt.title('Correlation Heatmap of Iris Dataset Features', fontsize=16)
plt.savefig('iris_correlation_heatmap.png', bbox_inches='tight')
```

Correlation Heatmap of Iris Dataset Features

